

22/7/26

H.W Java generates hash code for every object. Find out what is the hash code in Java and the purpose of generating hash code.

class:

H.W What is a Hash code in Java?

A hash code is a 32-bit integer value generated for every Java object. It is returned by the `hashCode()` method, which is defined in the object class.

By default, each object gets a hash code based on its identity. Classes like `String`, `Integer`, etc, override `hashCode()` so that objects with the same content produce the same hash code.

Purpose of Generating a Hash code.

The main purpose of a hash code are:

- Fast searching and retrieval of objects in hash-based collections such as `HashMap`, `HashSet` and `HashTable`.
- Efficient comparison of objects before calling the `equals()` method.
- Storing objects in hash tables by determining the bucket where an object should be placed.
- Improving performance of collection operations like insert, search, and delete.

Example Programs-

```
public class Main {  
    public static void main (String [] args) {  
        String s = "Anits",  
        System.out.println("Hash code: " + s.hashCode());  
        Object obj = new Object();  
        System.out.println("Object Hash code: " + obj.hashCode());  
    }  
}
```

Sample Output:

Hash Code: 63414081 // (may vary depending on the string)

Object Hash Code: 468121027 // (varies every execution).

Conclusion:

A hash code is a unique integer used to identify objects efficiently. It is mainly used by Java's hash-based collections to provide fast storage and retrieval of data.